Recruitment Matching Analysis

Optimized Numerical and Semantic Machine Learning | Internship Technical Report

1. Problem Statement

Recruiters compare applicant profiles with job requirements to identify suitable candidates. This study builds and evaluates machine learning classifiers that predict whether a recruitment-applicant pair is **matched (TARGET = 1)** or **unmatched (TARGET = 0)**. The models support initial analysis and are not intended to replace human recruitment decisions.

The study includes **Task 1**, which uses 10 numerical matching attributes, and **Task 2**, which adds six semantic-similarity features derived from Japanese text. Four classification algorithms are examined, with positive-class F1-score used as the principal performance measure.

2. Dataset Description

The provided synthetic dataset contains **3,000 recruitment-applicant pairs**. Each record includes identifiers, numerical matching scores, three text fields and a binary target. The target distribution is highly imbalanced:

Target class	Meaning	Records	Share
0	Unmatched	2,850	95%
1	Matched	150	5%

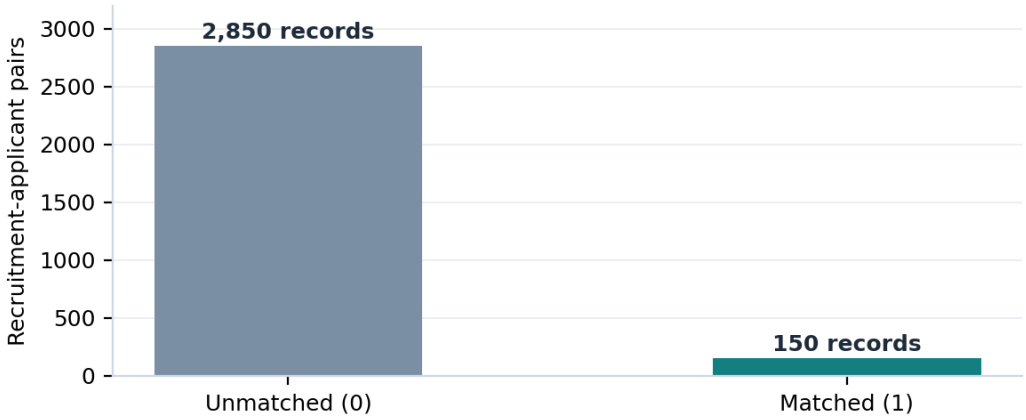


Figure 1. Distribution of the original target values.

> **Why imbalance matters:** Predicting every pair as unmatched would yield 95% accuracy while identifying no real matches. Therefore, precision, recall and F1-score are more informative than accuracy alone.

3. Data Preprocessing

Preprocessing was applied consistently so that the models could use the numerical input data without including target labels or identifiers as predictive features. The train/test split was performed **before** resampling or any training-dependent transformation.

3.1 Data cleaning and feature selection

Data audit: Check column names, data types, class labels, duplicates and missing values. The supplied numerical data contains no missing entries, but median imputation is retained inside the model pipeline as a safeguard. Japanese feature names are translated into working English labels without changing their values.

Feature selection: Only the ten numerical columns below are used in Task 1. Recruitment_ID, Applicant_ID, text columns and TARGET are excluded. Task 2 retains these ten inputs and adds six engineered semantic features.

Numerical feature group A	Numerical feature group B
Communication skill score	Job required-criteria match ratio
Applicant required-criteria match ratio	Job-experience BM25 score
Job/work-history Jaccard score	Total job-history count
Job-history count (last 3 years)	Job changes per career year
Job experience match score	Skill match score

3.2 Missing values and scaling

The pipeline applies **median imputation** to numerical attributes. **StandardScaler** is applied only for Logistic Regression, using the training-fold mean and standard deviation:

$$\text{Scaled value} = (\text{feature value} - \text{training mean}) / \text{training standard deviation}$$

Random Forest, Gradient Boosting and LightGBM use the imputed numerical features without standardization. Preprocessing parameters are learned from each training fold, never from held-out validation or test records.

3.3 Splitting strategy

Experiment	Training partition	Held-out test partition
Task 1 (latest)	2,400 rows (2,280 negative; 120 positive)	600 rows (570 negative; 30 positive)
Task 2 (earlier grouped)	2,424 rows (2,304 negative; 120 positive)	576 rows (546 negative; 30 positive)

> **Important distinction:** The latest Task 1 notebook uses a stratified row-level split; the earlier Task 2 experiment uses Recruitment_ID-disjoint groups. Both can be reported, but their headline test scores are not a controlled head-to-head comparison.

4. Handling Class Imbalance with Oversampling

Oversampling is used to reduce the influence of the 95:5 class imbalance during training. The notebook uses **RandomOverSampler**, which randomly duplicates existing matched-class training examples. It does not generate new synthetic feature values. For example, 1:1 oversampling duplicates the 120 positive examples until there are 2,280 matched training rows (2,160 additional duplicate rows).

4.1 What changes during oversampling?

In the latest Task 1 training partition there are 2,280 unmatched and 120 matched examples. The optimization searches three possible training configurations:

Balancing strategy	Unmatched (0)	Matched (1)	Positive:negative ratio
No oversampling	2,280	120	Approximately 1:19
Random oversampling 1:2	2,280	1,140	1:2
Random oversampling 1:1	2,280	2,280	1:1

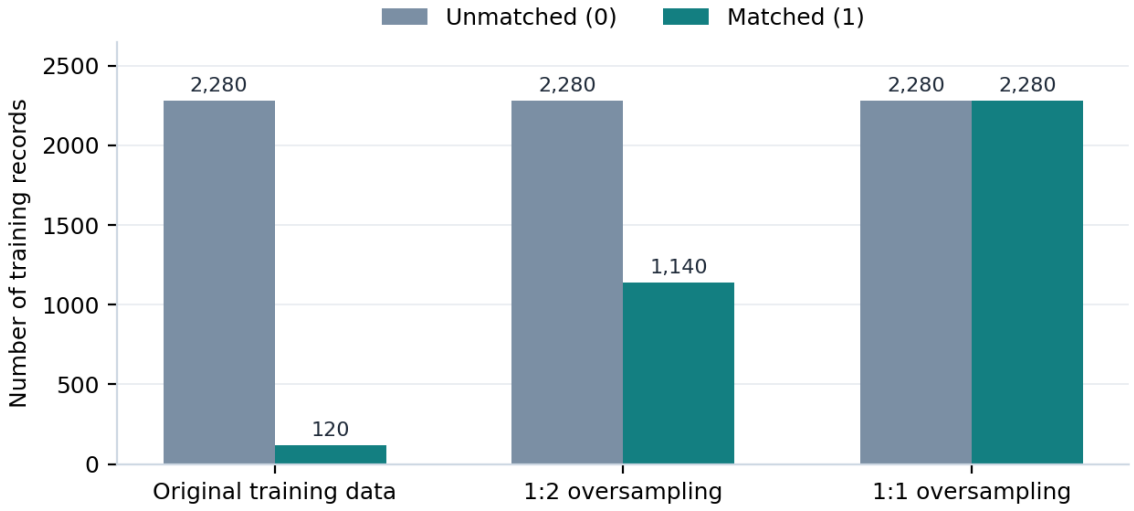


Figure 2. Class counts before and after random oversampling, illustrated for the full Task 1 training partition.

4.2 Where is oversampling applied?

Sampling is placed **inside the imbalanced-learn training pipeline** and applied separately within each cross-validation training fold. For example, a typical five-fold Task 1 training split contains 1,824 unmatched and 96 matched cases. At 1:1, only that fold is resampled to 1,824 of each class; the 480 validation rows remain unchanged.

01 Training fold	**02** Impute / scale	**03** Oversample positives	**04** Fit classifier

Leakage prevention: Never oversample the complete dataset before train/test splitting. Neither validation nor test data is oversampled. The Task 1 baseline fixes the ratio at 1:1; the optimized search chooses the balancing strategy independently for each model.

5. Algorithms and Optimization Method

The following binary classifiers were trained on identical feature sets within each respective task:

Algorithm	How it works / settings optimized
Logistic Regression	Linear probability model; regularization strength and penalty. Numerical inputs are standardized.
Random Forest	Combines many decision trees; searches tree count, depth, sample requirements and feature sampling.
Gradient Boosting	Adds trees sequentially to correct errors; searches learning rate, tree count, depth and subsampling.
LightGBM	Gradient-boosted decision trees; searches learning rate, leaves, depth and regularization.

5.1 Five-fold cross-validation

The training data is divided into five parts. In each round, four parts are used to fit preprocessing, perform optional oversampling and train the model; the remaining part is used for validation. Each part serves as the validation fold exactly once.

01 Training portion	**02** Five CV rounds	**03** Record F1 per fold	**04** Select configuration

5.2 Hyperparameter search and threshold optimization

For each Task 1 classifier, the optimized notebook tests **8 sampled hyperparameter configurations**, **3 balancing strategies** and **5 folds**, resulting in **120 training fits per algorithm** (480 across four algorithms). A classification threshold between 0.05 and 0.95 is selected from training-only out-of-fold predictions to maximize positive-class F1.

The selected configuration is then fitted to the full training partition and evaluated once on the held-out test partition. The earlier grouped Task 2 experiment also optimized parameters, balancing and thresholds, but its search size and group-aware splitting were different.

5.3 Evaluation metrics

Metric	Simple definition
Precision	TP / (TP + FP): fraction of predicted matches that are correct.
Recall	TP / (TP + FN): fraction of real matches that are detected.
F1-score	2 x Precision x Recall / (Precision + Recall): balances precision and recall.
ROC-AUC	Measures ranking quality over all possible probability thresholds.

Optimization objective: Choose model settings from training validation results, not from test labels. A higher validation score does not guarantee an improved score on unseen records.

6. Task 1: Results Using Ten Numerical Features

Task 1 compares baseline and optimized models on the same 600-row, stratified test partition. The baseline uses 1:1 RandomOverSampler and a fixed probability threshold of 0.50. The optimized models use training-selected parameters, balancing strategies and thresholds.

Algorithm	Baseline F1	Optimized F1	Optimized precision	Optimized recall
Logistic Regression	0.6374	0.8667	0.8667	0.8667
Random Forest	0.7500	0.8276	0.8571	0.8000
Gradient Boosting	0.7606	0.8308	0.7714	0.9000
LightGBM	0.7647	0.8000	0.8800	0.7333

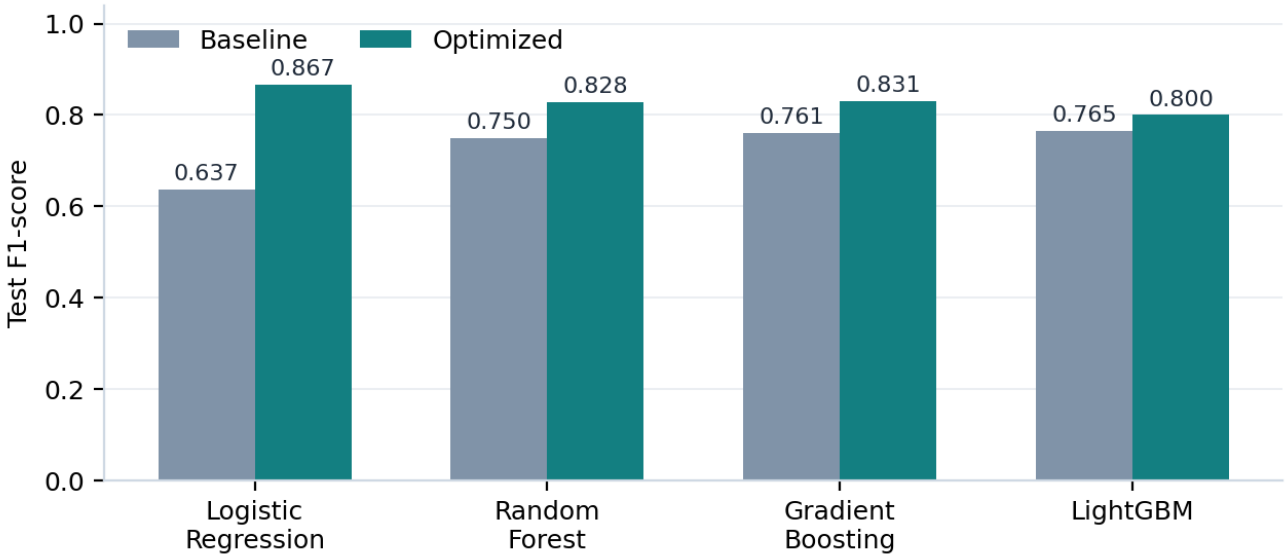


Figure 3. Task 1 baseline versus optimized F1-score on the identical held-out partition.

6.1 Selected settings and observed improvement

Algorithm	Training balancing	Threshold	F1 change (percentage points)
Logistic Regression	1:1 oversampling	0.92	+22.93
Random Forest	No oversampling	0.51	+7.76
Gradient Boosting	1:2 oversampling	0.77	+7.02
LightGBM	No oversampling	0.50	+3.53

Main observation: The highest observed Task 1 test F1 was **0.8667 (86.67%)** for optimized Logistic Regression. In contrast, Random Forest had the highest training-only selected out-of-fold F1 (~0.7845). The best observed test result should therefore be treated as exploratory, not as a new test-based model-selection rule.

7. Task 1: Logistic Regression Diagnostics

The optimized Logistic Regression classifier used L2 regularization (C approximately 0.1065), 1:1 training-only oversampling and a decision threshold of 0.92. Its final test precision, recall and F1 were each **0.8667**.

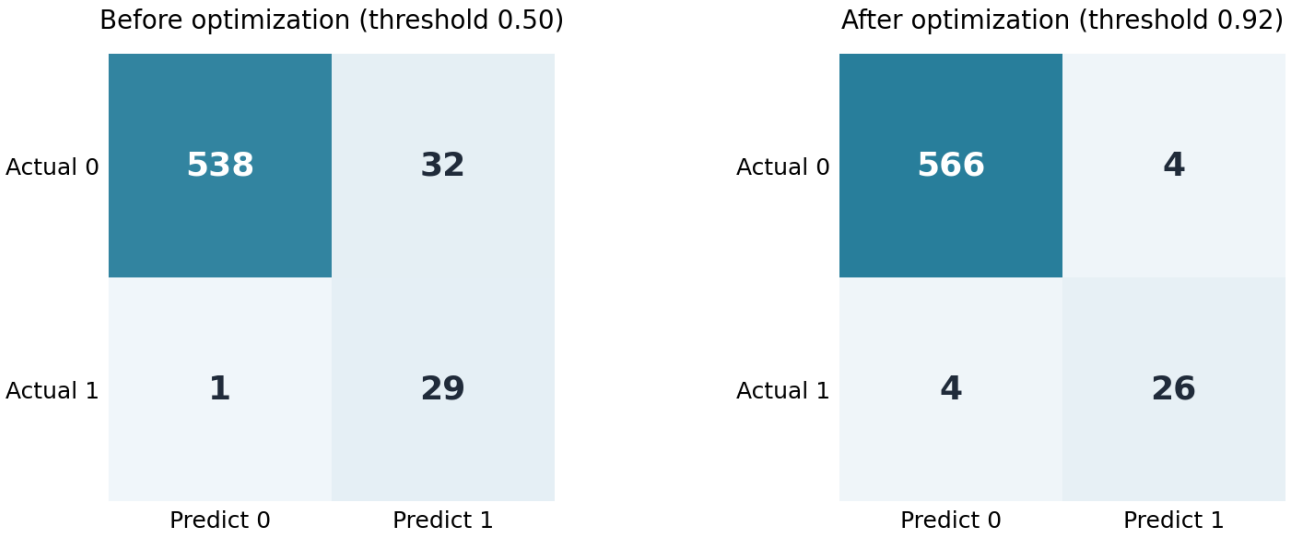


Figure 4. Confusion matrices for Logistic Regression before and after optimization.

Before optimization, the model correctly identified 29 positive matches but produced 32 false positives. After optimization, it identified 26 positive matches with only 4 false positives, while false negatives rose from 1 to 4. The higher threshold reduced incorrect match recommendations at the cost of missing three additional matches.

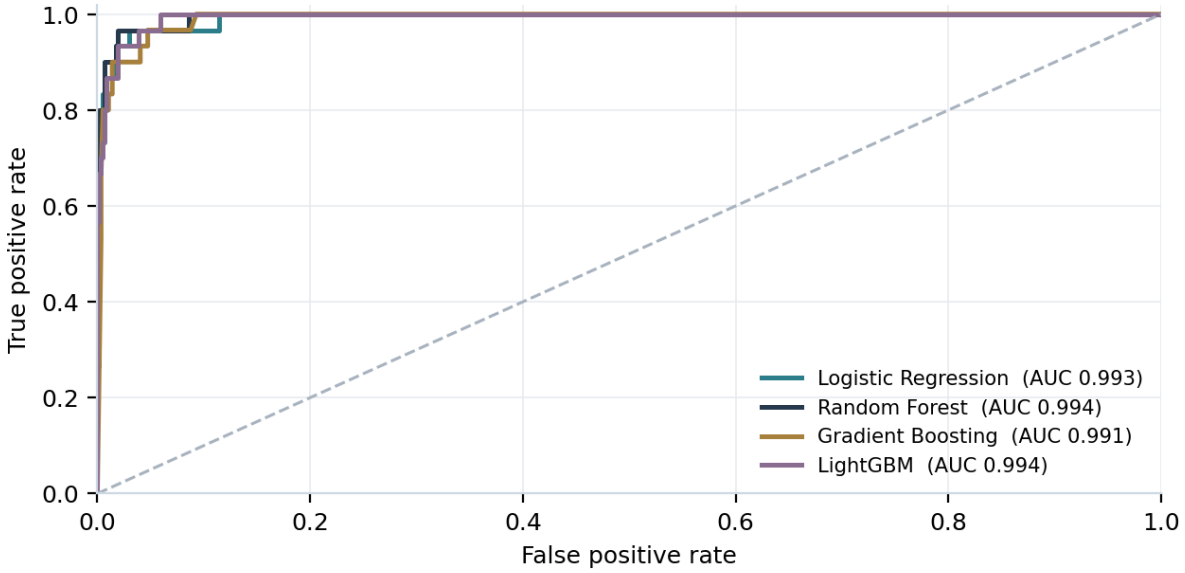


Figure 5. ROC curves for the four optimized Task 1 algorithms.

The optimized models all achieved ROC-AUC values near 0.99 on this synthetic test partition. ROC-AUC summarizes ranking quality, while F1 captures the precision-recall trade-off at the model's selected operating threshold.

8. Task 1: Feature Analysis and Interpretation

For Logistic Regression, signed coefficients show the direction of association with the predicted probability of a match after scaling. The chart reports the optimized coefficients. A larger absolute coefficient indicates a greater effect on the model score per standardized feature change, holding other inputs fixed; it does not establish causation.

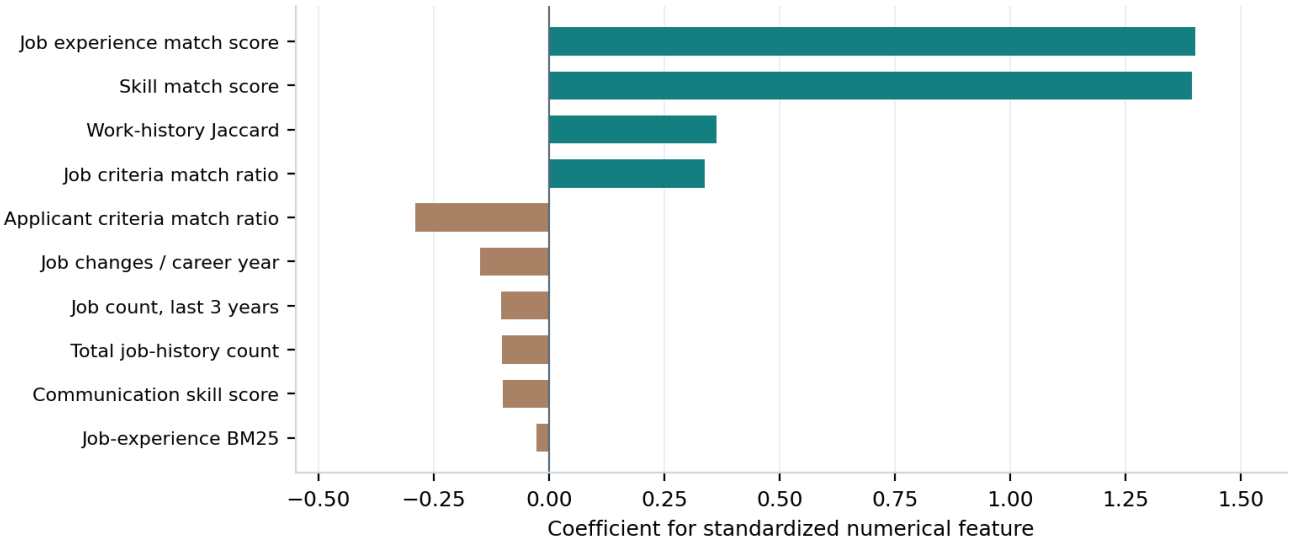


Figure 6. Signed feature coefficients of optimized Logistic Regression (positive and negative associations).

8.1 Principal observations

- **Job experience match score** and **skill match score** have the largest positive standardized coefficients (approximately 1.40 each).
- Work-history Jaccard and required-criteria matching ratios contribute additional predictive signals.
- Negative coefficients for recent job counts and career-change rate are model associations, not evidence that these attributes should determine employment decisions.

8.2 Important dataset limitations

The input dataset is synthetic. Engineered features such as JOB_EXP_MATCH_SCORE and SKILL_MATCH_SCORE may overlap with the mechanism used to construct the target labels; this requires checking the data-generation logic. Furthermore, repeated recruitment and applicant identities cross the row-level Task 1 split. Independent group-disjoint and real-world testing would be necessary before any operational deployment.

> **Interpretation:** The 86.67% result is an observed holdout F1 in the supplied synthetic dataset. It should not be described as verified performance for unseen jobs or previously unseen applicants.

9. Task 2: Semantic Similarity Feature Engineering

Task 2 extends the numerical matching approach by capturing the meaning of Japanese-language text. Three original fields are used: **Job**, **Choice Point** and **Skills**. The previously reported Task 2 experiment used a frozen pretrained **paraphrase-multilingual-MiniLM-L12-v2** encoder to generate one 384-dimensional embedding for each text field. The encoder itself was not fine-tuned.

9.1 Text-to-feature pipeline

01 Three text fields	**02** MiniLM embeddings	**03** Pairwise similarities	**04** Six numeric values

The embedding captures features of text meaning. Similarity is measured using cosine similarity, which compares the direction of two numerical embedding vectors:

Cosine similarity(A, B) = (A dot B) / (length(A) x length(B))

Three pairwise cosine scores are computed, followed by the mean, maximum and minimum of those three scores:

Additional semantic feature	Definition
Similarity 1	Job text and Skills text
Similarity 2	Job text and Choice Point text
Similarity 3	Choice Point text and Skills text
Similarity 4	Mean of Similarities 1-3
Similarity 5	Maximum of Similarities 1-3
Similarity 6	Minimum of Similarities 1-3

9.2 Final model inputs and preprocessing

Each Task 2 record contains **10 numerical attributes + 6 scalar semantic similarity features = 16 features**. The raw 384-dimensional embedding vectors are not concatenated directly with the tabular data. Numerical features are imputed as necessary; Logistic Regression inputs are standardized. All preprocessing and class balancing are fitted only on training folds.

The earlier Task 2 experiment used a recruitment-group-disjoint test partition (2,424 training and 576 test records) and grouped training cross-validation. This is distinct from the latest row-stratified Task 1 experiment.

> **What is being optimized?** Task 2 optimizes the downstream Logistic Regression, Random Forest, Gradient Boosting and LightGBM classifiers. The pretrained multilingual MiniLM weights remain unchanged.

10. Task 2: Grouped Test Results

The earlier executed Task 2 experiment was evaluated on 576 test records (546 unmatched and 30 matched), with Recruitment_ID groups separated from training. Four optimized classifiers were compared. The values below are from the earlier execution logs, **not** from the newer Task 2 Colab notebook, which has not yet been run with real MiniLM features.

Algorithm	Test F1	Precision	Recall
Logistic Regression	0.8077	0.9545	0.7000
Random Forest	0.8333	0.8333	0.8333
Gradient Boosting	0.7925	0.9130	0.7000
LightGBM	0.8364	0.9200	0.7667

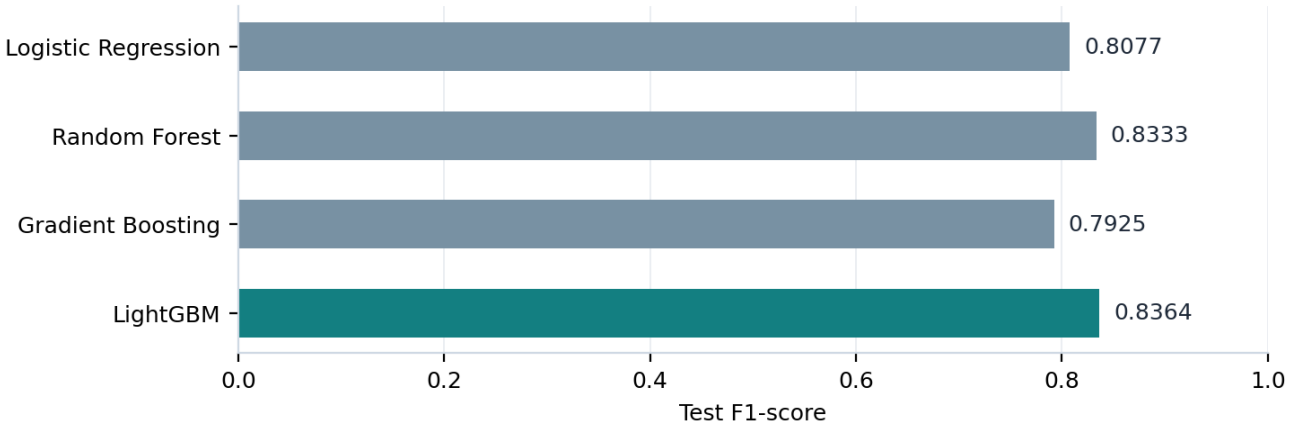


Figure 7. Earlier Task 2 group-disjoint test F1 by algorithm.

10.1 Best Task 2 classifier: optimized LightGBM

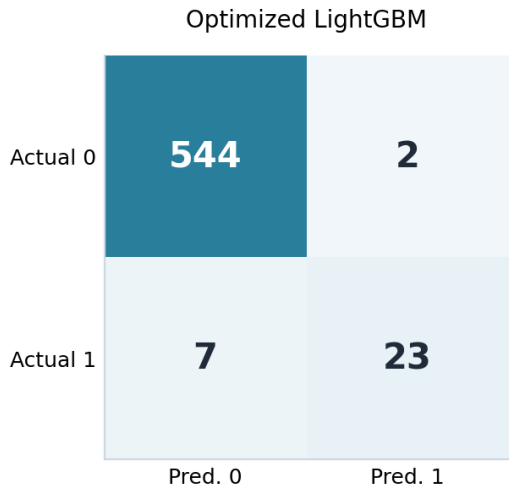


Figure 8. Confusion matrix for the earlier Task 2 LightGBM model.

The model used a **1:2 positive-to-negative training oversampling ratio** and a selected threshold of **0.73**. It correctly identified 23 positive cases, with 2 false positives and 7 false negatives. Precision was 92.00%, recall was 76.67% and F1 was **83.64%**.

$$F1 = (2 \times 23) / (2 \times 23 + 2 + 7) = 0.8364$$

11. Comparison, Limitations and Conclusion

The headline Task 1 (86.67%) and Task 2 (83.64%) scores were obtained on **different test partitions**. They should not be interpreted as proof that numerical inputs outperform semantic features. For a more meaningful assessment, the earlier experiment also evaluated both tasks on the same 576-record group-disjoint test partition.

Model	Task 1: numerical F1	Task 2: numerical + semantic F1
Logistic Regression	0.8077	0.8077
Random Forest	0.8462	0.8333
Gradient Boosting	0.8077	0.7925
LightGBM	0.8364	0.8364

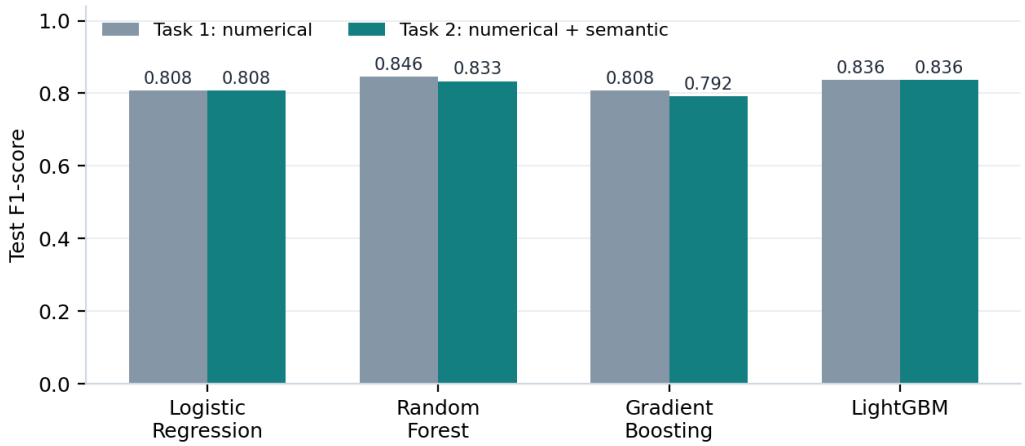


Figure 9. Same-grouped-holdout comparison, using the earlier paired evaluation.

On this common holdout, the highest Task 1 test F1 was **84.62%** (Random Forest) and the highest Task 2 test F1 was **83.64%** (LightGBM). Adding the six semantic features did not consistently improve positive-class F1 across the four classifiers.

11.1 Study limitations and future work

- **Small minority-class sample:** Each held-out evaluation contains only 30 positive matches; a few classification errors change F1 substantially.
- **Possible synthetic-label leakage:** Verify how the target was generated relative to engineered match-score inputs.
- **Identity overlap:** The Task 1 row-level split can repeat recruitment and applicant IDs across partitions; the earlier grouped split separates recruitment IDs but not necessarily applicant IDs.
- **Selection uncertainty:** Multiple configurations and previously viewed test outcomes make the reported maxima exploratory. Confirm on new data with a preselected model.

11.2 Conclusion

Optimization through cross-validated hyperparameter search, training-only random oversampling and threshold selection improved the observed Task 1 numerical-model F1, with Logistic Regression reaching 86.67% on its row-stratified holdout. The earlier text-enhanced Task 2 approach reached 83.64% with LightGBM on a different, group-disjoint holdout. A final, unified group-disjoint benchmark is the appropriate next step for evaluating whether semantic features improve generalization.

Data sources: Uploaded synthetic dataset; executed Task 1 notebook with saved metrics and plots; earlier Task 2 execution logs. The attached sample report was used for layout only.